23007

C23-EC/ECII/CIOT-102

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BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

FIRST YEAR (COMMON) EXAMINATION

ENGINEERING MATHEMATICS—I

Time : 3 hours] [*Total Marks* : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. If $A = \{0,1,2\}$ and $f : A \to B$ is a function such that $f(x) - x^2 + x + 1$, then find range of *f*.

2. Resolve $\dfrac{1}{(x+1)(x+3)}$ into partial fractions.

3. If $A = \begin{bmatrix} 1 & -1 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 4 & -1 \end{bmatrix}$, then find $A + 2B$.

4. If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, then find AB.

5. Prove that $\dfrac{+}{\cos 16^\circ - \sin 16^\circ} = \tan 61^\circ$.

6. Prove that $\dfrac{\sin 2}{1 - \cos 2A} = \quad A$.

7. Find the multiplicative inverse of the complex number $(4 + 3i)$.

8. Find the equation of the straight line passing through the point (1,2) and parallel to the line $x + y + 2 = 0$.

9. Evaluate $\lim_{x \to 0} \dfrac{\sin 7x}{\sin 3x}$.

10. Find the derivative of $2x^3 + 3x + 1$ w.r.t. x.

PART—B

10×5=50

Instructions : (1) Answer *any* **five** questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. Solve the following equations by using Cramer's rule : $x + y - z = 0$, $2x + y - z = 1$ and $3x + 2y + 2z = 5$.

12. *(a)* Prove that $\cos A + \cos(120° + A) + \cos(120° - A) = 0$.

(b) Prove that $\tan^{-1}\left(\dfrac{1}{4}\right) + \tan^{-1}\left(\dfrac{3}{5}\right) = \dfrac{\pi}{4}$.

13. *(a)* Solve : $\cos^2\theta + 2\cos\theta + 1 = 0$.

(b) Solve : ΔABC with $a = 8$, $b = 4$, $c = 4\sqrt{3}$.

14. *(a)* Find the equation of the circle having (3,4) and (1,2) as end points of its diameter.

(b) Find the equation of rectangular hyperbola whose focus is the point (1,–2) and directrix is the line $x + y + 5 = 0$.

15. *(a)* Find the derivative of $\sin x+\cos x+\tan x$ with respect to x.

(b) Find the derivative of x^2e^x with respect to x.

16. *(a)* If $y=(\sin x)^x$, then find $\frac{dy}{dx}$.

(b) If $u(x,y)=\log(x+y)$, then prove that $x\frac{\partial y}{\partial x}+y\frac{\partial y}{\partial y}=1$.

17. *(a)* Find the slopes of the tangent and normal to the curve $y=2x^2+3x+1$ at $(0,1)$.

(b) If $s(t)=16t+4t^2-t^3$, is a displacement of a particle, then find its velocity and acceleration at $t=1\,\text{sec}$.

18. *(a)* Find the maximum or minimum value of the function $f(x)=-x^2+6x+40$.

(b) If an error of 2% is made in the radius of a circle, then find the percentage error in measuring its area.

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